

KUNAL JOSHI

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Enterprise Security Solutions | Software Development | Cloud Governance & Automation

Experienced Security Engineer and AWS Cloud Security Specialist with 8 years of experience in designing, securing, and optimizing enterprise-grade applications. Expertise in developing next-generation technology solutions, implementing cryptographic protocols, and performing penetration testing to secure sensitive data. Passionate about delivering secure, high-performance software solutions that drive business success. Skilled in mentoring professionals on secure coding and encryption best practices.

AREAS OF EXPERTISE

- Software Development & Scripting
- Data Protection & Encryption
- Identity and Access Management
- AWS Security Architecture Design
- Cloud Security
- Infrastructure as Code (IaC)

TECHNICAL PROFICIENCIES

Development: Java, Python, C/C++ | **IAM:** OAuth, SAML, AWS IAM | **Cryptography:** AWS Key Management Service, HSM, PKCS#11, JCE, OpenSSL Dyn, Encryption SDK | **Security Operations:** Cloud SIEM and Security Operations Center, Threat hunting | Cloud Security, Cloud Development, Java Development, AWS Automation

HONORS & ACHIEVEMENTS

ACM ICPC 2016-Kolkata finals Rank 58 | Qualified - Google Code Jam (2017-18)
#2223 - Round 1 Code Jam (2017) | AIR 98 - Tata Codevita (2016)

EDUCATION & CERTIFICATES

B. Tech. in Computer Science College of Technology and Engineering – Udaipur, India (2017)
AWS Certified Security Specialty | AWS IAM SME | AWS Cloud HSM SME | AWS KMS SME | AWS Certified AI Practitioner

PROFESSIONAL EXPERIENCE

CLOUD SECURITY ARCHITECT | Amazon Web Services – Mumbai, India May 2024 – Present

- **Transformed security requirements into robust AWS cloud architectures integrated with AWS security services** like AWS IAM (Identity and Access Management), KMS, CloudTrail, and Guard Duty to enhance threat intelligence and access control measures.
- Delivered projects to customers to strengthen their **Identity and Access Management (IAM) posture** by implementing IAM best practices, such as the principle of least privilege, multi-factor authentication, credential rotation, Single Sign-On through identity providers such as Azure AD/Okta, and Just-In-Time (JIT) access.
- **Engineered and implemented end-to-end AWS security solutions for enterprise customers**, ensuring compliance with cloud security principles and safeguarding sensitive financial data in accordance with Indian regulatory standards.
- **Architected and deployed robust security architecture and action plans**, fortifying AWS security pillars such as identity and access management, logging and monitoring, incident response, data protection, and infrastructure security.
- **Designed and executed scalable security solutions**, integrating automated least privilege policies, patch management, IaC pipelines, Cloud Native SIEM using OpenSearch, and security scanners to enhance compliance and threat mitigation across multiple customer environments.

CLOUD ENGINEER II – SECURITY | Amazon Web Services – Bangalore, India Dec 2017 – May 2024

- **Augmented security operations by identifying and resolving complex issues across AWS services**, improving access control (IAM, Cognito, SSO), encryption management (KMS, Cloud HSM), and threat detection (GuardDuty, Inspector).
- **Designed AWS security training frameworks** covering identity and access management, Cloud HSM, Single Sign-On (SSO), and secrets manager, enabling internal teams to implement robust authentication and encryption strategies.
- **Recognized as a Subject Matter Expert in AWS IAM services**, providing deep expertise in Federation, SAML, and SSO implementations.
- Educated and empowered AWS clients by **publishing knowledge center content on IAM tag-based restrictions**. [Link](#).

- **Collaborated with cross-functional engineering teams to identify and fix SDK issues**, enhancing cross-language compatibility and improving performance across Python, Java, C++, C#, JavaScript, Ruby, Go, and PHP.
- **Expedited cloud adoption** by seamlessly onboarding **60+** applications into AWS Single Sign-On Cloud Application Catalog.
- **Served as a Cryptography Subject Matter Expert (SME), specializing in AWS KMS and Cloud HSM**, resolving escalations and optimizing cryptographic operations using Java JCE, PKCS#11, and OpenSSL dynamic engine.

PROJECTS

Cloud Native SIEM and SOC (For a leading stock brokerage firm)

- Engineered and implemented Cloud native SIEM (Security Information and Event Management) using AWS OpenSearch.
- Developed custom log processor in Java, with the complete system capable of processing 10+ million events per minute.
- Integrated public and privately subscribed threat intelligence with the log ingestion pipeline.
- Developed custom logic to integrate AWS Incident Manager with AWS OpenSearch for incident-case management and paging alerts.
- Achieved near real time alerting (from log generation) and visualization for log volume of ~1TB per day without downtimes.

KMS Key Policy – Least privilege orchestrator (For a leading multi-national bank)

- Engineered cloud-based solution to identify and remediate excessive cryptographic key privileges, supporting PCI-DSS and SOC 2 compliance.
- Developed analytics engine processing high-volume CloudTrail logs to identify key usage patterns and recommend least-privilege policies.
- Built automated GitHub integration with approval workflows to visualize, track, and audit policy changes across thousands of KMS keys.
- Implemented comprehensive alerting and reporting for smooth user experience, improving compliance and security posture.

IAM Policy - Least Privilege Orchestrator with Compliance Automation (For a leading multi-national bank)

- Engineered a full-stack IAM governance platform to identify and remediate excessive permissions for IAM users and roles across a complete AWS Organization, supporting SOC 2 and ISO 27001 compliance requirements.
- Automated access reviews and entitlement management by analyzing organization-wide AWS CloudTrail logs to identify usage patterns and recommend least-privilege policies.
- Integrated SAML-based authentication with AWS Identity Center and implemented 2PR approval workflow for policy remediation, ensuring segregation of duties and audit trail compliance.
- Built AWS CloudScape design-based UI with role-based access control (RBAC) for security teams and administrators.
- Improved security posture by remediating excessive permissions for thousands of IAM users/roles across hundreds of AWS accounts, reducing attack surface.
- Generated automated compliance reports for quarterly access audits, reducing manual effort.

AWS CloudHSM Key Management Dashboard (personal)

- Developed a dashboard lets one configure CloudHSM connection through a web interface, then manage cryptographic keys with a clean UI.
- Allows creating AES, RSA, and EC keys, search through existing keys, and delete keys when needed.
- Built with React on the frontend and FastAPI on the backend, using PyKCS11 to connect directly to CloudHSM cluster. Uses SQLite for session data storage.
- https://github.com/joshikunal94/cloudhsm_management_dashboard